



Maciej Karczmarz

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📍 Stalowa Wola, Poland

📁 [Portfolio Page](#)

🐙 [Github](#)

🌐 [LinkedIn](#)

🎨 [ArtStation](#)

Projects

-----Programming Projects-----

Gix-> Developed a real-time currency exchange rate tool that aggregates data from local exchange offices for accurate/up-to-date currency tracking.

SmartOffer-> Automation system for AB Bechcicki. It transforms inquiries into ready-to-use cost estimates (Premium/Budget) by selecting materials based on technical standards and parameters, not just brands. Built on n8n workflows, and AWS, ensuring enterprise-grade scalability.

Nequ3D-> Spatial data pipeline and desktop analytics platform that converts raw LiDAR and photogrammetry datasets into optimized OpenUSD scenes via NVIDIA RTX NTC. Features an asynchronous Go backend, containerized Python workers, a Wails frontend with local LLM reporting, and a GitOps-driven Kubernetes architecture.

GRATS-> Created a minimalist graphics renderer to provide a learning platform for OpenGL and graphics programming fundamentals.

Sand-Simulation-> Granular physics engine focused on realistic particle behavior. The project simulates natural environmental interactions, emphasizing efficient computation and physical accuracy.

-----Graphics Projects-----

Unreal Engine 5 Project-> Developing realistic environments using Gaea 2.0 and Houdini tools, with focus on procedural workflows and software integration.

ArtStation Profile-> Created a variety of 3D assets and visualizations, ranging from realistic architectural renders and product visualizations to stylized fantasy scenes.

Languages

Polish

Native

English

C1

About me

I am interested in building practical tools, API integrations and improving them. I have experience with Go, Python, n8n, AWS and data processing. I enjoy finding repetitive and inefficient processes and turning them into simple solutions. I work well at the intersection of technology and real business needs, I learn new tools quickly and I am able to clearly explain complex issues. My experience in backend development, 3D graphics and real-time systems helps me solve problems efficiently and prototype quickly.

Experience

Simplicity Games

Rzeszów, Poland

Technical Artist

November 2024 - January 2025

3D Model Creation & Optimization – designing models in Blender from scratch and based on reference materials, creating optimized low-poly meshes, LODs, and collision meshes. Composing and processing PBR textures using tools such as Substance Painter, baking normal maps. Optimizing assets for performance and compliance with Unity Engine requirements, including UV improvements.

<https://simplicitygames.pl/>

Education

Rzeszow University of Technology

October 2022 - February 2026

Computer Science

Bachelor's Studies – Faculty of Electrical and Computer Engineering

Engaged in a variety of academic and additional projects combining software development, data analysis, web technologies and 3D graphics. Key activities and achievements include:

- **C++ development** – designed and implemented a graphical simulation project as part of coursework.
- **Data Analysis** – applied Python, MATLAB, and R for data processing, visualization, and statistical modeling.
- **Team-based Web Development** – collaborated with peers from the faculty on a group project using React and modern web technologies.
- **AI Research Project** – developed an experimental AI solution (MLP Algorithm variants) within the Artificial Intelligence course, exploring machine learning concepts.
- **Hackathon Participation** – took part in events such as NASA Space Apps Challenge, DIGIEDUHACK, Hack SPACESHIELD or Hack CARPATHIA, focusing on rapid prototyping and problem-solving under time constraints.
- **Student Research Group** – member of SKNI KOD since 2022, contributing to multiple collaborative and independent projects in software engineering and 3D graphics.

Rzeszow University of Technology March 2026 - Present
Computer Science

Master's Studies – Faculty of Electrical and Computer Engineering